

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – VI

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Date of Assignment: 21/08/2026	Date/Time of Submission:21/08/2026

Aim:Designing Effective and Visually Appealing Dashboards

a) Apply worksheet-level and workbook-level formatting

Task 1: Connect to the Dataset and Build a Baseline Chart

1. Open Tableau Desktop.
2. Click Text File under Connect.
3. Select your CSV dataset and click Open.
4. Open Sheet 1.
5. Rename the sheet as Category Profitability.
6. Drag Category to the Rows shelf.
7. Drag Profit to the Columns shelf.

Task 2: Apply Workbook-Level Formatting

1. Click Format → Workbook.
2. Under Workbook Fonts, select Segoe UI (or Tableau Book).
3. Set Worksheet Titles to Segoe UI Semibold, 14 pt.
4. Scroll to Lines.
5. Turn Grid Lines Off (or Light Gray).
6. Close the Format pane.

Task 3: Apply Worksheet-Level Formatting

1. Right-click anywhere on the chart.
2. Select Format.

Format Fonts

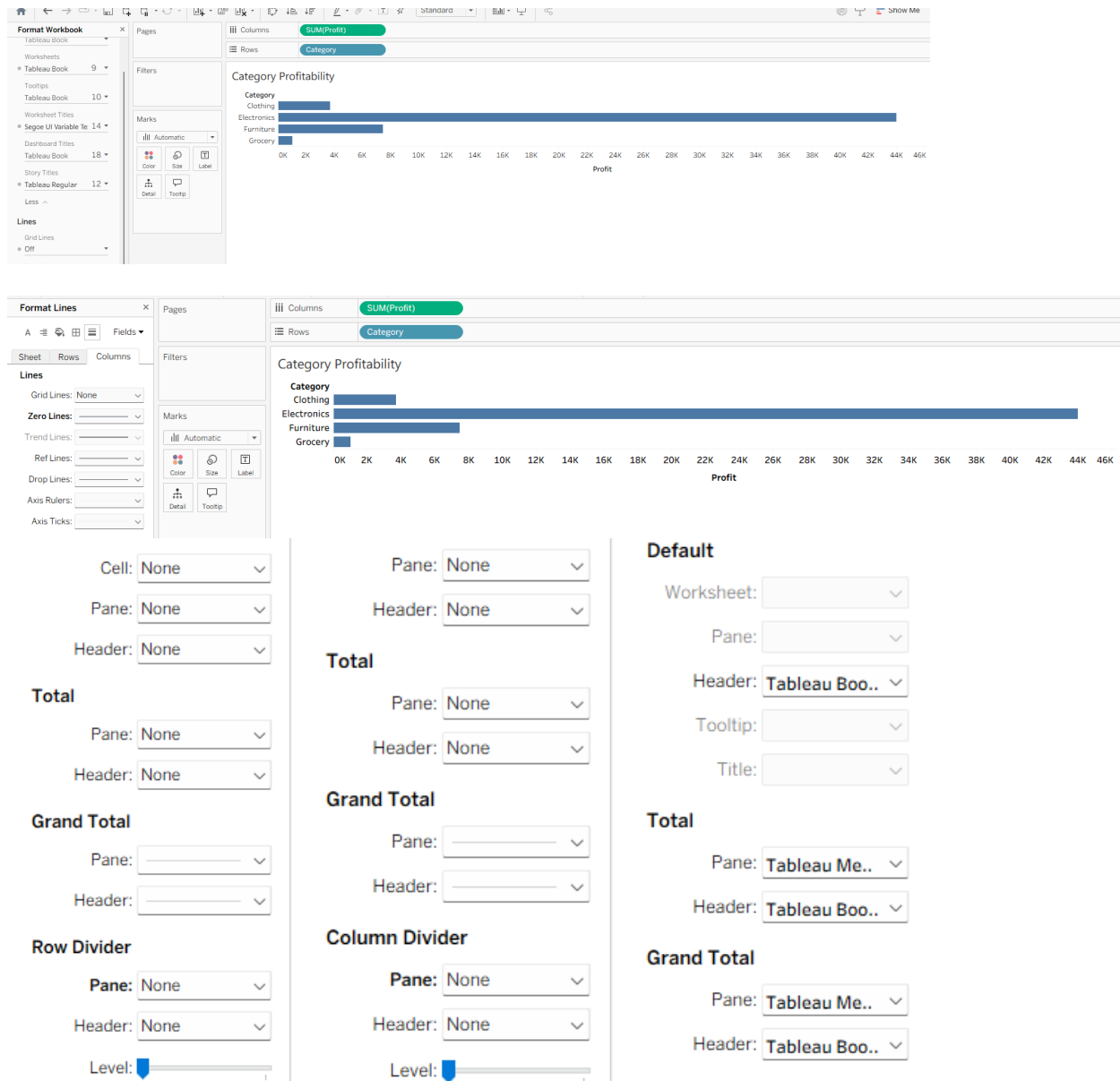
- Header → Bold, 10 pt
- Pane → Regular, 10 pt

Format Borders

- Row Divider → None
- Column Divider → None

Format Zero Lines

- Go to Lines.
- Set Zero Lines → Solid Dark Gray.



b) Customize number, date, and null value formatting

Task 1: Format Amount and Profit as Currency

1. Right-click Profit.
2. Select Default Properties → Number Format.
3. Choose Currency (Custom).
4. Set Prefix to ₹.
5. Decimal Places = 0.
6. Click OK.
7. Repeat the same for Amount.

Task 2: Format Order Date

1. Right-click Order Date.
2. Select Default Properties → Date Format.

3. Select Custom.
4. Enter:
dd-MMM-yyyy
5. Click OK.

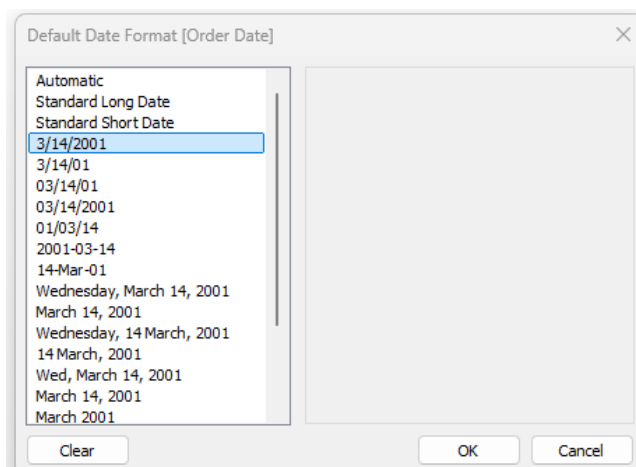
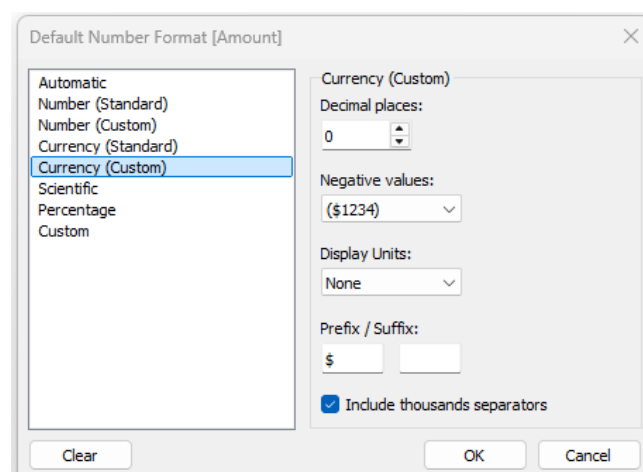
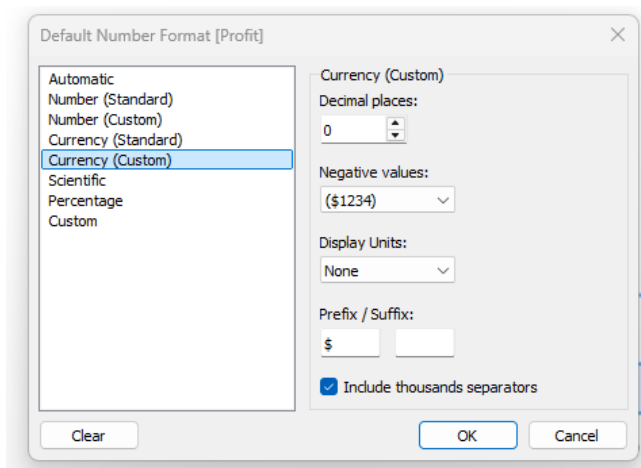
Task 3: Handle Null Values

1. Create a new worksheet.
2. Rename it City Sales & Nulls.
3. Drag City to Rows.
4. Drag Amount to Columns.
5. Right-click SUM(Amount).
6. Select Format.
7. Scroll to Special Values (NULL).
8. Replace NULL text with:

N/A

or

Not Reported



Special Values (eg. NULL)

Text: Not Reported

Marks: Show at Indi... ▾

c) Design effective color schemes and layouts

Task 1: Apply Profit Color Scheme

1. Open Category Profitability.
2. Drag Profit to Color.
3. Click Color → Edit Colors.
4. Select Orange-Blue Diverging.
5. Enable Stepped Color.
6. Set Center = 0.
7. Click Apply → OK.

Task 2: Create KPI Summary

1. Create a new worksheet.
2. Rename it KPI Summary.
3. Change Marks to Text.
4. Drag Amount to Text.
5. Drag Profit to Text.
6. Drag Order ID to Text.
7. Change Order ID to Count (Distinct).

Arrange the text as:

TOTAL SALES

<SUM(Amount)>

TOTAL PROFIT

<SUM(Profit)>

TOTAL ORDERS

<CNTD(Order ID)>

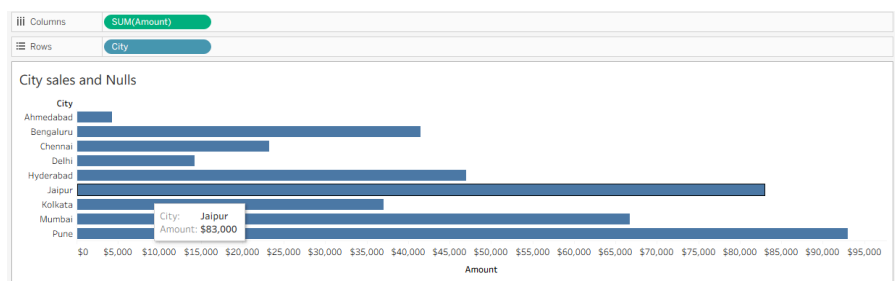
8. Make metric values Bold, 18 pt.
9. Make headings Bold, 9 pt.
10. Set Alignment → Center.

Task 3: Create Dashboard Layout

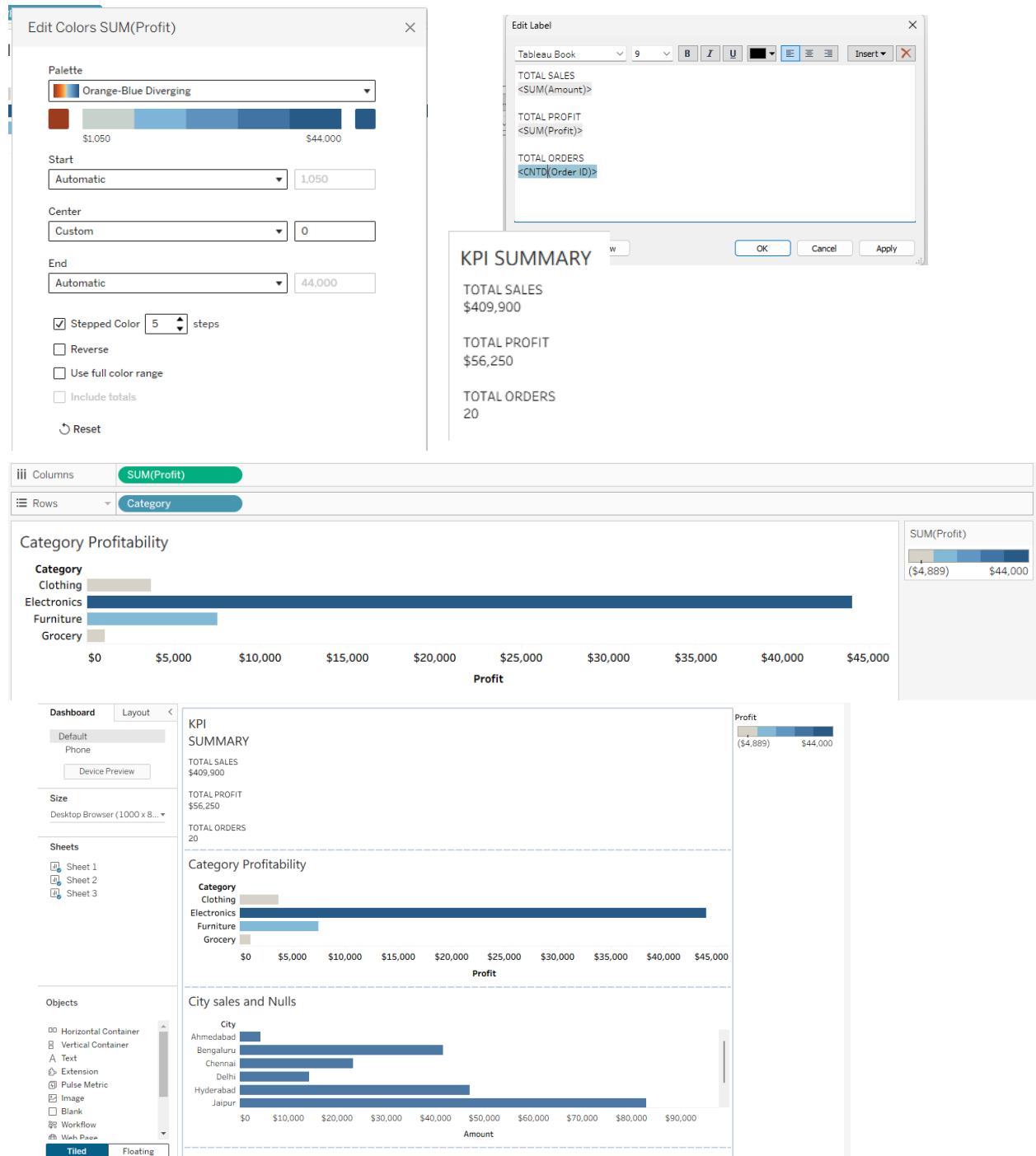
1. Create a New Dashboard.
2. Rename it:

Executive Sales Overview

3. Dashboard Size → 1000 × 800.
4. Choose Tiled layout.
5. Drag a Vertical Container.
6. Add:
 - KPI Summary
 - Category Profitability
 - City Sales & Nulls



7. Hide unnecessary worksheet titles if required.



d) Create informative tooltips

Task 1

1. Open Category Profitability.
2. Click Tooltip.

Task 2

Drag these fields to Tooltip:

- Amount
- Quantity
- Order ID (Count Distinct)

Task 3

Replace the default tooltip with:

Category Overview

Generated <SUM(Amount)> in total sales across

<CNTD(Order ID)> orders,

selling <SUM(Quantity)> items.

Net Profit:

<SUM(Profit)>

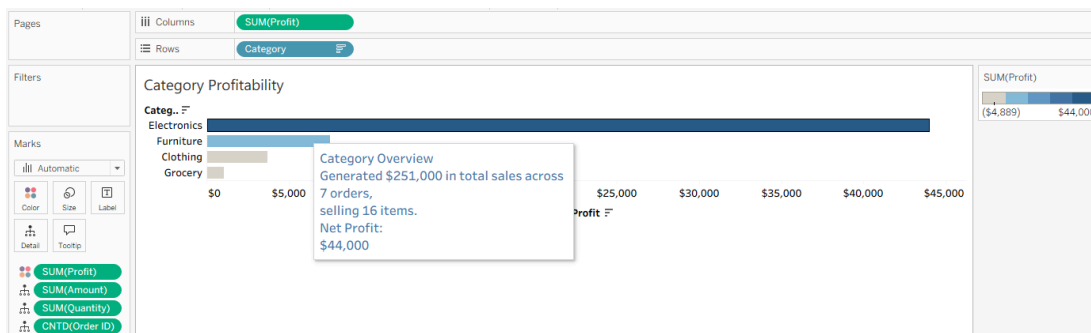
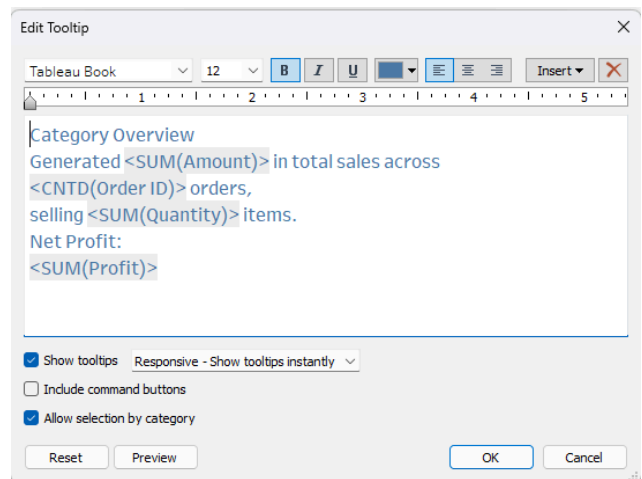
Make:

- Title → Bold, 12 pt
- Metrics → Bold
- Body → Regular, 10 pt

Disable Include Command Buttons.

Task 4

Hover over every bar and verify the tooltip.



e) Implement “Viz in Tooltip”

Task 1

1. Create a worksheet.
2. Rename it:

Tooltip SubCategory Detail

3. Drag Sub-Category → Rows.
4. Drag Profit → Columns.
5. Drag Profit → Color.
6. Sort Descending.
7. Change Fit → Entire View.

Task 2

1. Return to Category Profitability.
2. Open Tooltip.

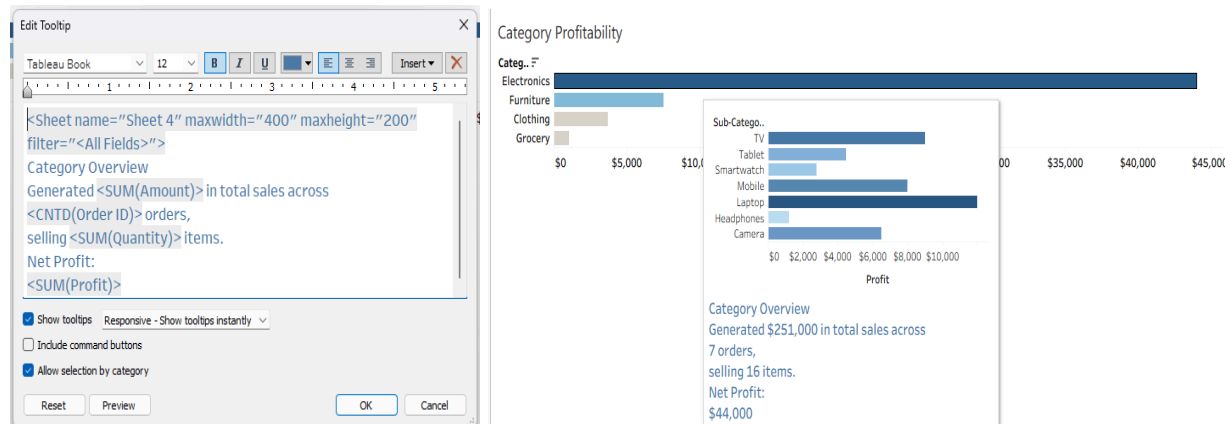
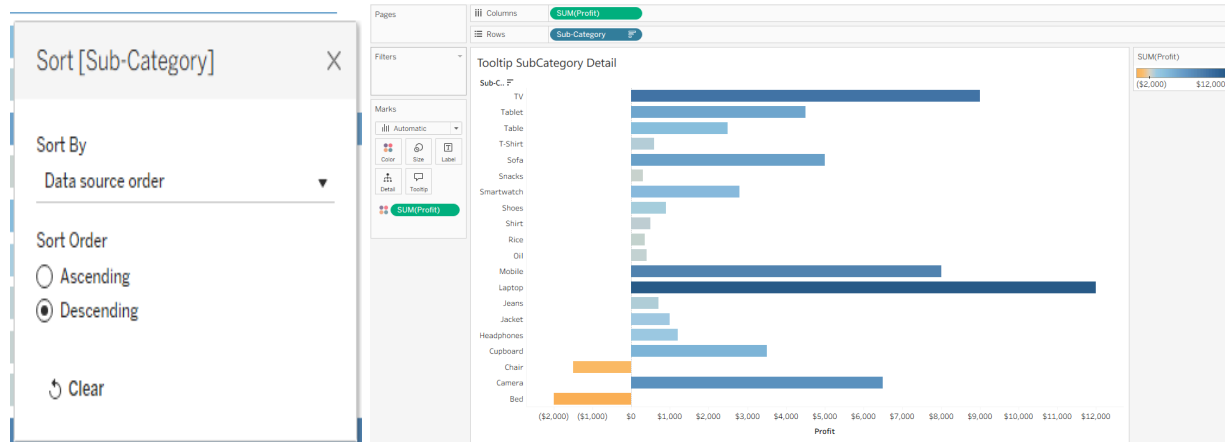
- Click Insert → Sheets.
- Select:

Tooltip SubCategory Detail

- Set:
 - Width = 400
 - Height = 200

Task 3

Hover over any category and verify that the Sub-Category chart appears inside the tooltip.



f) Improve Readability and User Experience

Task 1: Improve Dashboard Layout

- Open the Executive Sales Overview dashboard.
- Select each worksheet on the dashboard one by one.
- Resize the worksheets to ensure all charts are clearly visible.
- Arrange the worksheets neatly to maintain proper spacing and alignment.
- If available, adjust the Outer Padding from the Layout pane; otherwise, manually leave sufficient spacing between the charts.
- Ensure the dashboard is easy to read and all worksheet titles are visible.

Task 2: Create a Highlight Action

- Open the dashboard.

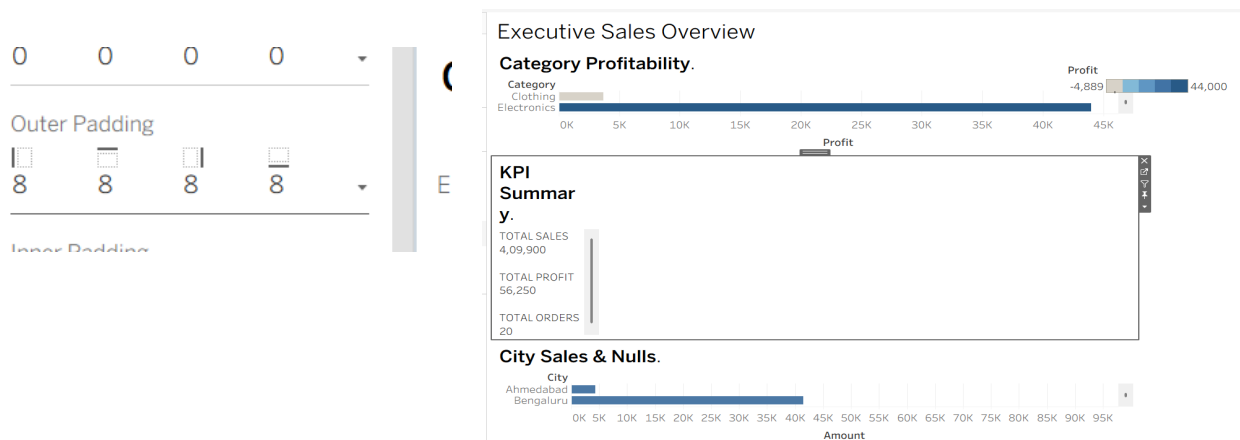
2. From the menu bar, select Dashboard → Actions.
3. Click Add Action → Highlight.
4. Enter the action name as Highlight Category.
5. Under Source Sheets, select Category Profitability.
6. Under Run Action On, select Hover.
7. Under Target Sheets, select All Worksheets.
8. Leave Target Highlighting as All Fields.
9. Click OK, then click OK again to save the action.

Task 3: Remove Unnecessary Legends

1. Click the Color Legend displayed on the dashboard.
2. Click the More Options (▼) arrow on the legend.
3. Select Hide Card if the legend is not required.
4. If you want to keep the legend, select Floating and move it to a suitable location.
5. Repeat the process for any other unnecessary legends or filter cards.
6. Verify that the dashboard appears clean and uncluttered.

Task 4: Verify Dashboard Interactivity

1. Open the dashboard in Presentation Mode (or press F5/F7, depending on your Tableau version).
2. Hover over the Category Profitability chart to verify the Highlight Action.
3. Hover over the chart to verify that the customized Tooltip is displayed.
4. Check that the dashboard fonts, colors, and worksheet alignment are consistent.
5. Confirm that all dashboard components are displayed correctly.
6. Exit Presentation Mode and save the workbook.



g) Apply best practices for visual storytelling

Task 1: Arrange the Dashboard

1. Open the Executive Sales Overview dashboard.
2. Place the KPI Summary worksheet at the top of the dashboard.
3. Position the Category Profitability worksheet below the KPI Summary.
4. Place the City Sales & Nulls worksheet below or beside the Category Profitability chart.

- Adjust the size of each worksheet so the dashboard presents the information in a logical sequence.

Task 2: Rename Worksheet Titles

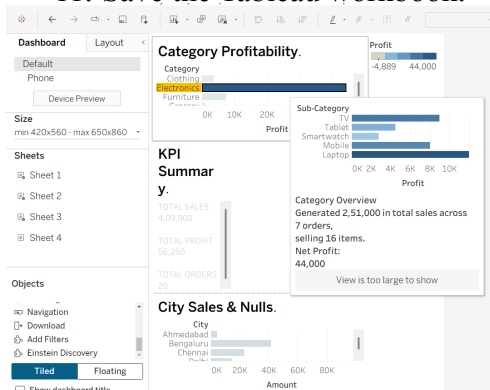
- Double-click the title of the Category Profitability worksheet.
- Rename it to a meaningful title, such as:
Technology & Office Supplies Lead Profitability
- Double-click the title of the City Sales & Nulls worksheet.
- Rename it to:
Top Cities Generate Maximum Revenue
- Click outside the title area to save the changes.

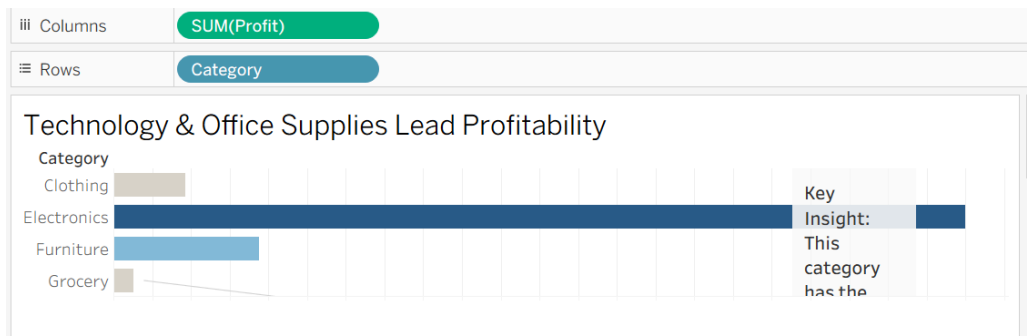
Task 3: Add an Annotation

- Open the Category Profitability worksheet.
- Right-click the bar representing the lowest profit category.
- Select Annotate → Mark.
- Enter the following annotation:
Key Insight:
High discount rates in this category reduce profit margins.
- Click OK.
- Drag the annotation box to a suitable position beside the selected bar.

Task 4: Create a Story

- Click the New Story icon at the bottom of Tableau.
- Rename the story to E-Commerce Executive Presentation.
- Set the Story Size to Automatic.
- Drag the Executive Sales Overview dashboard onto the story canvas.
- In the caption box, enter:
Executive Summary: Sales remain strong across top categories.
- Click Blank to create Story Point 2.
- Drag the dashboard onto the second story point.
- Highlight the category with the lowest profit.
- Update the caption with:
Profit Bottleneck: Furniture margins are dragged down by shipping costs and return rates.
- Review both story points to ensure they present the dashboard in a logical sequence.
- Save the Tableau workbook.





E-Commerce Executive Presentation

Executive Summary: Sales remain strong across the major product categories with healthy overall profitability.

Profit Bottleneck: This category generates the lowest profit and should be reviewed for pricing.

Executive Sales Overview

KPI Summary

TOTAL REVENUE
4,09,900

TOTAL PROFIT
56,250

TOTAL ORDERS
20

Profit

-4,889 44,000

Technology & Office Supplies Lead Profitability



Top Cities Generate Maximum Revenue

